

JEE-Main-04-04-2026 (Memory Based)
[MORNING SHIFT]
Physics

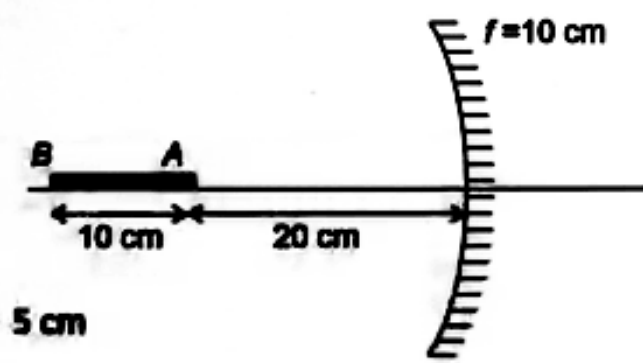
Question: Radius of a soap bubble is increased from 1 cm to 2 cm. Work done in process is (S is surface tension)

Options:

- (a) $\pi S \times 10^{-2}$ J
- (b) $1.2 \pi S$ J
- (c) $2.4 \pi S \times 10^{-3}$ J
- (d) $\pi S \times 10^{-3}$ J

Answer: (c)

Question: A rod is placed along principal axis as shown. Find length of image.



Options:

- (a) 10 cm
- (b) 5 cm
- (c) 15 cm
- (d) 20 cm

Answer: (b)

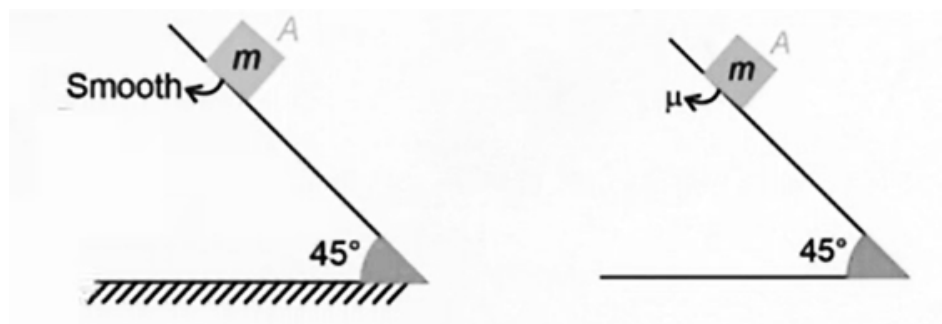
Question: Nuclei A and B form a nucleus C. BE/N for A, B and C are 3 MeV, 7 MeV and 6 MeV. Then energy produced in $2A^3 + B^4 \rightarrow C^{10}$

Options:

- (a) 8 MeV
- (b) 12 MeV
- (c) 14 MeV
- (d) 10 MeV

Answer: (c)

Question: Find coefficient of friction if time taken by block in rough surface is 50% more than time taken by smooth surface. The distance slide by the mass is same in both the cases.



Options:

(a) $\mu = 5/9$

(b) $\mu = 4/9$

(c) $\mu = 5/7$

(d) $\mu = 4/7$

Answer: (b)

Question: Inputs of A and B are given. Find output result Y of the circuit



Options:

(a) $A + \bar{B}$

(b) $\bar{A} + B$

(c) $\bar{A} B$

(d) $A. \bar{B}$

Answer: (a)

Question: A uniform sphere of mass m and radius R is divided into two equal parts one part of sphere has moment inertia I_1 about diameter and moment of inertia of disc of

$$\frac{l_2}{2}$$

radius 2R and mass m about perpendicular axis through its center is I_2 . Find $\frac{l_2}{I_1}$.

Options:

(a) 10

(b) 2

(c) 5

(d) 1

Answer: (a)

Question: If the wavelength of first line of Balmer series and of Brackett series of H are

$$\frac{\lambda_1}{\lambda_2}$$

respectively λ_1 and λ_2 respectively, then value of $\frac{\lambda_1}{\lambda_2}$ is

Options:

- (a) 0.16
- (b) 0.04
- (c) 0.28
- (d) 0.35

Answer: (a)

Question: A slit of width a is illuminated by the light of wavelength λ . The linear separation between 1st and 3rd minima in the diffraction pattern produced on a screen placed at a distance D from the slit is ____.

Options:

- (a) $3D\lambda$
- (b) $D\lambda$
- (c) $1.5D\lambda/a$
- (d) $2D\lambda/a$

Answer: (d)

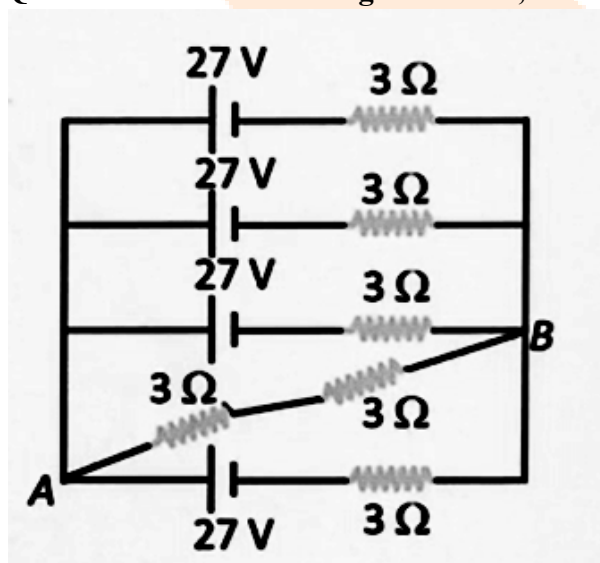
Question: In a screw gauge when the circular scale is given five complete rotations it increases linearly by 2.5mm. If the circular scale has 100 divisions, the least count of screw gauge is _____ mm.

Options:

- (a) 1×10^{-2}
- (b) 5×10^{-3}
- (c) 1×10^{-3}
- (d) 5×10^{-2}

Answer: (b)

Question: For the circuit given below, find the V_{AB} and i_{AB} .



Options:

- (a) 18 V, 8 A

- (b) 18 V, 6 A
- (c) 24 V, 3 A
- (d) 24 V, 4 A

Answer: (d)

Question: The increase in the pressure required to decrease the volume (ΔV) of water is $6.3 \times 10^7 \text{ N/m}^2$. The percentage decrease in the volume is ____
(Bulk modulus of water = $2.1 \times 10^9 \text{ N/m}^2$)

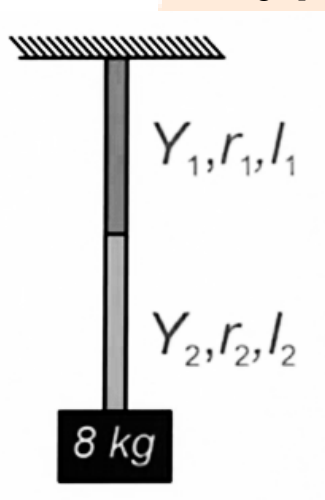
Options:

- (a) 3
- (b) 2
- (c) 0
- (d) 1

Answer: (a)

Question: A system of two composite wire is having a mass 8 kg attached with it as shown in figure.

For wire (1) $Y_1 = 2 \times 10^{11} \text{ N/m}^2$, $r_1 = 2 \text{ mm}$ and $l_1 = 0.314 \text{ m}$. And for wire (2) $Y_2 = 4 \times 10^{11} \text{ N/m}^2$, $r_2 = 2 \text{ mm}$ and $l_2 = 0.628 \text{ m}$. Find total extension in the wires. [$Y \rightarrow$ Young modulus, $r \rightarrow$ Radius, $l \rightarrow$ Length]



Options:

- (a) $4 \times 10^{-5} \text{ m}$
- (b) $4 \times 10^{-4} \text{ m}$
- (c) $2 \times 10^{-5} \text{ m}$
- (d) $2 \times 10^{-4} \text{ m}$

Answer: (c)

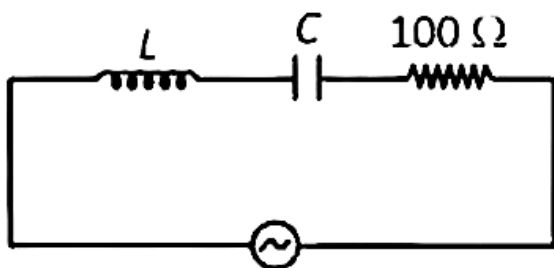
Question: A spiral coil having total 200 turns, inner radius 3 cm and outer radius 6 cm has 20 mA current through it. Find its magnetic moment.

Options:

- (a) $42\pi \times 10^{-4} \text{ Am}^2$
- (b) $36\pi \times 10^{-3} \text{ Am}^2$
- (c) $84\pi \times 10^{-4} \text{ Am}^2$
- (d) $27\pi \times 10^{-4} \text{ Am}^2$

Answer: (c)

Question: Power factor of the below circuit is 0.5. Find $|X_L - X_C|$. (take $\sqrt{3} = 1.73$)



Options:

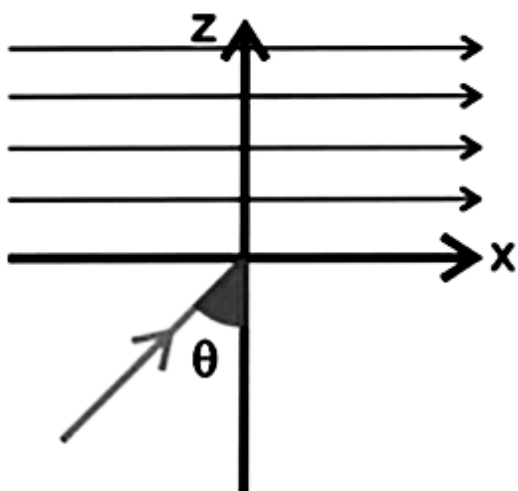
- (a) 346 Ω
- (b) 173 Ω
- (c) 86.5 Ω
- (d) 58 Ω

Answer: (b)

Question: Electric field of EMW is given as

$$\vec{E} = (E_1 \hat{i} + E_2 \hat{k}) \sin(kx + kz - \omega t).$$

Find reflected ray equation if θ is Brewster angle.



Options:

- (a) $(E_1 \hat{i} + E_2 \hat{k}) \sin(kx + kz - \omega t)$
- (b) $E_1 \hat{i} + E_2 \hat{k} \sin(kx - kz - \omega t)$
- (c) $E_0 \hat{j} \sin(kx + kz - \omega t)$
- (d) $E_0 \hat{j} \sin(kx - kz - \omega t)$

Answer: (d)

Question: A telescope has resolving power of 5×10^{-7} radian. The wavelength of light is 500 nm. Find diameter of telescope. (in fermi meter)

Options:

- (a) 112
- (b) 224
- (c) 61
- (d) 305

Answer: (d)

Question: Two forces $\vec{F}_1$ and $\vec{F}_2$ act simultaneously on a body

$\vec{F}_1 = 3\hat{i} + 5\hat{j}$ and $\vec{F}_2 = 2\hat{i} - 4\hat{j} + 2\hat{k}$ and body is displaced by 25 cm along $3\hat{i} - 4\hat{j}$ find work done by net force

Options:

- (a) 55 J
- (b) 60 J
- (c) 70 J
- (d) 86 J

Answer: (a)

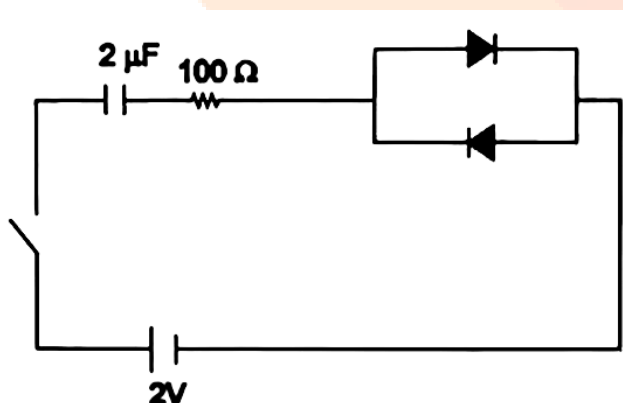
Question: Two projectiles are projected with the same initial velocities at 15° and 30° angles with respect to the horizontal. The ratio of the range is ratio 1 : X. The value of X is

Options:

- (a) 0
- (b) $\sqrt{3}$
- (c) 1
- (d) 4

Answer: (b)

Question: If resistance of diode in forward bias is 10Ω then time constant of the circuit would be $\alpha \times 10^{-5}$. Find α



Options:

- (a) 23×10^{-5}
- (b) 22×10^{-5}
- (c) 2×10^{-5}
- (d) 25×10^{-5}

Answer: (b)

JEE-Main-04-04-2026 (Memory Based)**[MORNING SHIFT]****Chemistry**

Question: Statement-I: BP order is $HI < HBr < HCl < HF$

Statement-II: MP order is $HI < HF < HCl < HBr$.

Options:

- (a) Statement-I and Statement-II are correct
- (b) Statement-I is correct and Statement-II is wrong
- (c) Statement-I is wrong and Statement-II is incorrect
- (d) Statement-I wrong and Statement-II is correct

Answer: (c)

Solution: (c) Statement-I is wrong and Statement-II is incorrect

Question: Given below are two statements:

Statement I: Maltose is a non-reducing sugar

Statement II: Lactose is a reducing sugar

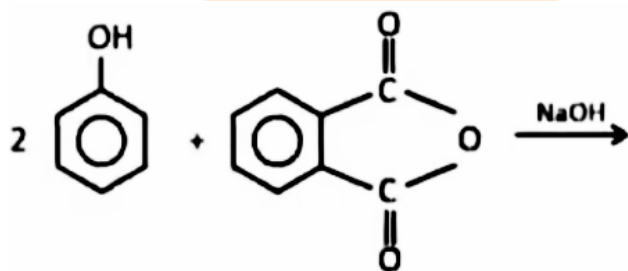
Options:

- (a) Both statement I and statement II are correct
- (b) Statement I is correct but statement II is incorrect
- (c) Statement I is incorrect but statement II is correct
- (d) Both statement I and statement II are incorrect

Answer: (c)

Solution: (c) Statement I is incorrect but statement II is correct

Question: Consider the following reaction



The color of the compound formed is

Options:

- (a) Violet
- (b) Red
- (c) Colorless
- (d) Pink

Answer: (d)

Solution: (d) Pink

Question: If the wavelength of first line of Balmer series and of Brackett series of H are respectively λ_1 and λ_2 respectively, then value of $\frac{\lambda_1}{\lambda_2}$ is

Options:

- (a) 0.16
- (b) 0.04
- (c) 0.28
- (d) 0.35

Answer: (a)

Solution: (a) 0.16

Question: What will be the pH at 25°C of solution obtained by mixing of 100 mL, 0.5 M NH_4OH solution and 0.01 M, 25 mL NH_4Cl solution? (pK_b of $\text{NH}_3 = 4.74$)

Options:

- (a) 11.56
- (b) 9.0
- (c) 2.44
- (d) 4.5

Answer: (a)

Solution: (a) 11.56

Question: Match the following

Species	M
i) Mn^{2+}	P) $\sqrt{24}$
ii) Co^{2+}	Q) $\sqrt{35}$
iii) Cu^{2+}	R) $\sqrt{15}$
iv) Cr^{2+}	S) $\sqrt{3}$

Options:

- (a) i - Q, ii - R, iii - S, iv - P
- (b) i - R, ii - S, iii - P, iv - Q
- (c) i - P, ii - Q, iii - R, iv - S
- (d) i - S, ii - Q, iii - P, iv - R

Answer: (a)

Solution: (a) i - Q, ii - R, iii - S, iv - P

Question: Match the following

List-I (Amino Acid)	List-II (Single letter code)
i) Arginine	P) D
ii) Lysine	Q) K
iii) Glutamic acid	R) E
iv) Aspartic acid	S) R

Choose the correct option

Options:

(a) i - S, ii - Q, iii - P, iv - R

(b) i - Q, ii - S, iii - P, iv - R

(c) i - S, ii - Q, iii - R, iv - P

(d) i - P, ii - Q, iii - R, iv - S

Answer: (c)

Solution: (c) i - S, ii - Q, iii - R, iv - P

Question: Given

$A + B \rightleftharpoons C; K_{eq} = 2.7 \times 10^{-5}$

$\frac{1}{3}A + \frac{1}{3}B \rightarrow \frac{1}{3}C; K_{eq}^1 = x$

The value of x is

Options:

(a) 9×10^{-6}

(b) 3×10^{-2}

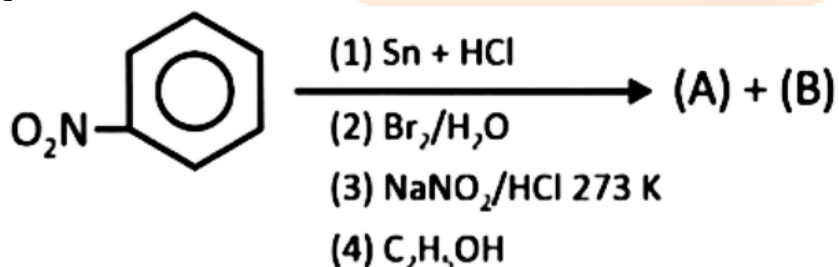
(c) 2.7×10^{-5}

(d) 4×10^{-2}

Answer: (b)

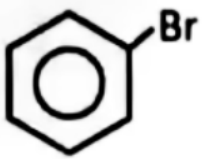
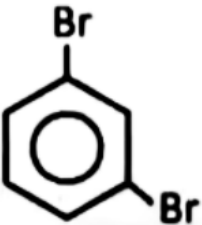
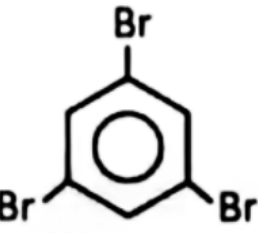
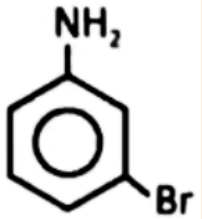
Solution: (b) 3×10^{-2}

Question: Consider the reaction:



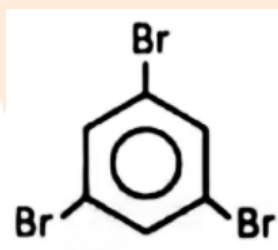
Identify 'A' (it is benzenoid product)

Options:

- (a) 
- (b) 
- (c) 
- (d) 

Answer: (c)

Solution: (c)



Question: If 'n' is the number of lone pairs of electrons in the equatorial position of the most stable structure of ClF_3 , which of the following ions also have 'n' unpaired electrons?

- A. V^{3+}
 B. Ti^{3+}
 C. Cu^{2+}
 D. Ni^{2+}
 E. Ti^{2+}

Options:

- (a) A and C only
 (b) A, D and E only
 (c) B and C only

(d) B and D only

Answer: (b)

Solution: (b) A, D and E only

Question: Consider the statements:

Statement I: Actinoid contraction > Lanthanoid contraction (element to element)

Statement II: Tb^{4+} , Ce^{4+} – oxidant

Statement III: Yb^{2+} , Eu^{2+} – reductant

Mark the correct statement(s):

Options:

(a) I, II only

(b) I, II only are correct

(c) I, II, III are correct

(d) Only II is correct

Answer: (c)

Solution: (c) I, II, III are correct

Question: Match List-I with List-II and choose the correct option.

List-I (Naming Reaction)	List-II (Reactions)
I) Wurtz reaction	P) $R-X + AgF \rightarrow R-F + AgX$
II) Swarts reaction	Q) $R-X + NaI \rightarrow R-I + NaX$
III) Finkelstein reaction	R) $2Ar-X + 2Na \rightarrow Ar-Ar + 2NaX$
IV) Fittig reaction	S) $2R-X + 2Na \rightarrow R-R + 2NaX$

Options:

(a) i - P, ii - Q, iii - R, iv - S

(b) i - S, ii - P, iii - Q, iv - R

(c) i - P, ii - S, iii - Q, iv - R

(d) i - R, ii - P, iii - Q, iv - S

Answer: (b)

Solution: (b) i - S, ii - P, iii - Q, iv - R

Question: The calculated spin-only magnetic moment of $K_3[Fe(OH)_6]$ and $K_4[Fe(OH)_6]$ respectively are:

Options:

(a) 5.92 BM and 4.90 BM

(b) 5.92 BM and 5.92 BM

(c) 4.90 BM and 4.90 BM

(d) 4.90 BM and 5.92 BM

Answer: (a)

Solution: (a) 5.92 BM and 4.90 BM

Question: Arrange the following functional groups in increasing order of electron withdrawing effect.

$-\text{CN}$; $-\text{NO}_2$; $-\text{I}$; $-\text{COOH}$

Options:

- (a) $-\text{NO}_2 < -\text{CN} < -\text{I} < -\text{COOH}$
- (b) $-\text{I} < -\text{COOH} < -\text{CN} < -\text{NO}_2$
- (c) $-\text{COOH} < -\text{I}, -\text{NO}_2 < -\text{CN}$
- (d) $-\text{I} < -\text{COOH} < -\text{NO}_2 < -\text{CN}$

Answer: (b)

Solution: (b) $-\text{I} < -\text{COOH} < -\text{CN} < -\text{NO}_2$

Question: The number of moles of FeC_2O_4 and $\text{FeSO}_4 \times (\text{NH}_4)_2\text{SO}_4 \times 6\text{H}_2\text{O}$ required to completely react with 2 mol of acidified KMnO_4 for each of them is respectively:

Options:

- (a) 3.33 moles and 10 moles
- (b) 5 moles and 10 moles
- (c) 5 moles and 3.33 moles
- (d) 3.33 moles and 5 moles

Answer: (a)

Solution: (a) 3.33 moles and 10 moles

Question: Consider the following reaction:

$\text{P} \rightarrow \text{Q}$

If the reaction follows first order kinetics, find the fraction of decomposed molecule at time t

Options:

- (a) $1 - e^{kt}$
- (b) $1 + e^{kt}$
- (c) $1 - e^{-kt}$
- (d) $1 + e^{-kt}$

Answer: (c)

Solution: (c) $1 - e^{-kt}$

Question: 4 gram of solute is dissolved in a certain amount of solvent and its vapour pressure decreases from 760 mm to 750 mm. 4 gram of same solute is dissolved in same amount of that solvent that causes increase in boiling point by 0.15°C . Calculate the molar mass of solute. ($K_b = 0.5 \text{ K kg/mol}$). Consider solute to be non-electrolyte.

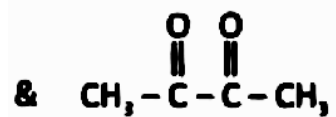
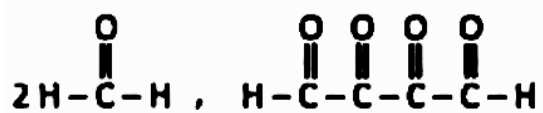
Options:

- (a) 22.22 g/mol
- (b) 44.44 g/mol
- (c) 60 g/mol
- (d) 70 g/mol

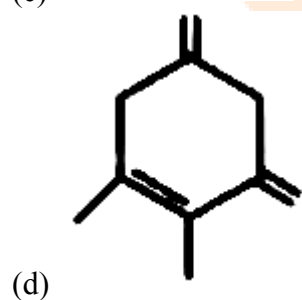
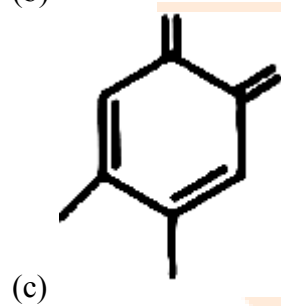
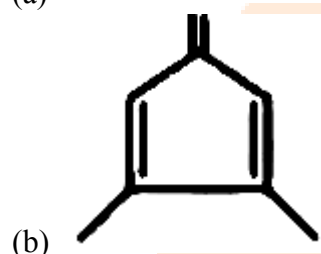
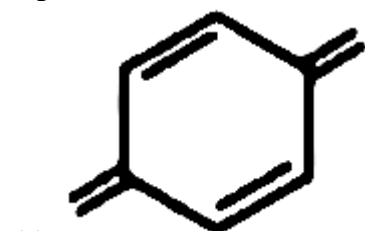
Answer: (b)

Solution: (b) 44.44 g/mol

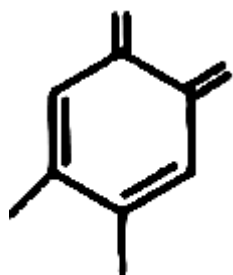
Question: An organic compound on reductive ozonolysis gives the following products



The structure of organic compound is
Options:



Answer: (c)



Solution: (c)

Question: 112 dm³ of SO₂ is given at STP. The number of moles and number of molecules of SO₂ respectively are:

Options:

- (a) 0.5, 30.11×10^{24}
- (b) 5, 3.011×10^{24}
- (c) 8, 3.011×10^{23}
- (d) 8, 6.022×10^{24}

Answer: (b)

Solution: (b) 5, 3.011×10^{24}

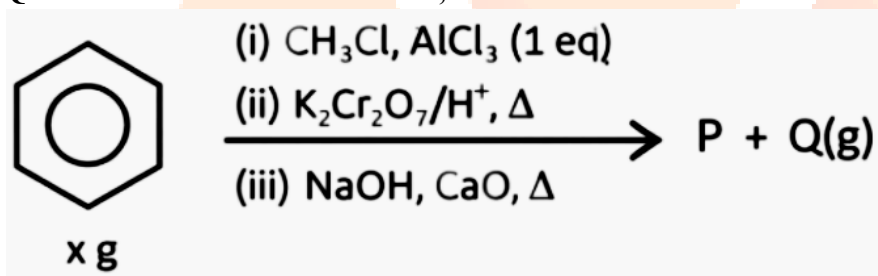
Question: Total number of lone pairs and σ bond pairs formed by central atom (Xe) in XeO₆⁴⁻

Options:

Answer: (6)

Solution:

Question: Consider the reaction,



Volume of Q(g) is 0.112 dm³ at STP. Find value of 10(x) (Nearest integer).

Options:

Answer: (4)

Solution:

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[MORNING SHIFT]
Maths

Question: Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{a, b, c\}$. Then number of functions which are not onto are

Answer: (93)

Solution:

Total number of functions from A to B = $|B|^{|A|} = 3^5 = 243$

Now, count onto (surjective) functions:

Using inclusion-exclusion:

Number of onto functions

$$= 3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5$$

$$= 243 - 3 \times 32 + 3 \times 1$$

$$= 243 - 96 + 3 = 150$$

So, number of functions which are NOT onto:

$$= 243 - 150 = 93$$

Question: Let A is a matrix of order 3 such that $|A| = -4$. Then, the value of $|\text{adj}(\text{adj}(2\text{adj}(A))^{-1})|$ is

Options:

(a) $\frac{1}{2^{16}}$

(b) $\frac{1}{2^{28}}$

(c) $\frac{1}{2^{30}}$

(d) $\frac{1}{2^{20}}$

Answer: (b)

Solution:

$$|\text{adj}(A)| = |A|^{n-1} = (-4)^2 = 16$$

$$|2 \text{adj}(A)| = 2^3 \cdot 16 = 8 \cdot 16 = 128$$

$$|\text{adj}(2 \text{adj}(A))| = 128^2 = 2^{14}$$

Apply adj again:

$$|\text{adj}(\text{adj}(2 \text{adj}(A)))| = (2^{14})^2 = 2^{28}$$

Question: The number of ways 4 boys and 3 girls are to be arranged in a row so that all 3 girls are not together, is equal to

Options:

- (a) 2320
- (b) 4320
- (c) 4920
- (d) 1440

Answer: (b)

Solution:

First, calculate the total number of ways to arrange all 7 children (4 boys and 3 girls) without any restrictions. This is a permutation of 7 distinct objects:

$$\text{Total ways} = 7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5,040$$

Now, you have 4 boys and 1 girl-block, which is $4 + 1 = 5$ units to arrange. These can be arranged in $5!$ ways.

Inside the block, the 3 girls can be arranged among themselves in $3!$ ways.

Multiply these to get the total "together" arrangements:

$$\text{Girls together} = 5! \times 3! = 120 \times 6 = 720$$

Required ways = Total ways – Girls together

$$\text{Required ways} = 5,040 - 720 = 4,320$$

Question: A line with direction ratios 1, -1, 2 intersects the lines $\frac{x}{2} = \frac{y}{3} = \frac{z+1}{3}$ and

$\frac{x+}{-1} = \frac{y-2}{1} = \frac{z}{4}$ at the points P and Q respectively. If the length of the line segment PQ

is α , then $225\alpha^2$ is equal to

Options:

- (a) 1104
- (b) 1204
- (c) 1024
- (d) 1014

Answer: (d)

Solution:

From the first two ratios:

$$2\lambda + \mu + 1 = -(3\lambda - \mu - 2)$$

$$2\lambda + \mu + 1 = -3\lambda + \mu + 2$$

$$5\lambda = 1 \implies \lambda = \frac{1}{5}$$

From the first and third ratios:

$$2(2\lambda + \mu + 1) = 3\lambda - 4\mu - 1$$

$$4\lambda + 2\mu + 2 = 3\lambda - 4\mu - 1$$

$$\lambda + 6\mu = -3$$

Substituting $\lambda = \frac{1}{5}$:

$$\frac{1}{5} + 6\mu = -3 \implies 6\mu = -3 - \frac{1}{5} = -\frac{16}{5}$$

$$\mu = -\frac{16}{30} = -\frac{8}{15}$$

Now find the components of the vector $\vec{PQ}$:

$$\bullet \quad x_2 - x_1 = 2\left(\frac{1}{5}\right) + \left(-\frac{8}{15}\right) + 1 = \frac{6 - 8 + 15}{15} = \frac{13}{15}$$

$$\bullet \quad y_2 - y_1 = 3\left(\frac{1}{5}\right) - \left(-\frac{8}{15}\right) - 2 = \frac{9 + 8 - 30}{15} = -\frac{13}{15}$$

$$\bullet \quad z_2 - z_1 = 3\left(\frac{1}{5}\right) - 4\left(-\frac{8}{15}\right) - 1 = \frac{9 + 32 - 15}{15} = \frac{26}{15}$$

The length α is:

$$\alpha^2 = \left(\frac{13}{15}\right)^2 + \left(-\frac{13}{15}\right)^2 + \left(\frac{26}{15}\right)^2$$

$$\alpha^2 = \frac{169 + 169 + 676}{225} = \frac{1014}{225}$$

$$225\alpha^2 = 225 \times \frac{1014}{225} = 1014$$

Question: If $|z + 2| = |z - 2|$ and $\text{Arg}\left|\frac{z-3}{z+i}\right| = \frac{\pi}{4}$ then $|z|^2$

Solution:

$$|z + 2| = |z - 2|$$

$$\sqrt{(x+2)^2 + y^2} = \sqrt{(x-2)^2 + y^2}$$

$$(x+2)^2 + y^2 = (x-2)^2 + y^2$$

$$x^2 + 4x + 4 = x^2 - 4x + 4 \implies 8x = 0 \implies x = 0$$

$$\text{Arg} \left| \frac{z-3}{z+i} \right| = \frac{\pi}{4}$$

$$\frac{iy-3}{iy+i} = \frac{-3+iy}{i(y+1)}$$

$$\frac{(-3+iy)(-i)}{i(y+1)(-i)} = \frac{3i+y}{y+1} = \left(\frac{y}{y+1} \right) + i \left(\frac{3}{y+1} \right)$$

$$\tan \left(\frac{\pi}{4} \right) = \frac{\text{Imaginary part}}{\text{Real part}}$$

$$1 = \frac{\frac{3}{y+1}}{\frac{y}{y+1}} \Rightarrow 1 = \frac{3}{y} \Rightarrow y = 3$$

$$|z| = \sqrt{0^2 + 3^2} = 3$$

$$|z|^2 = 3^2 = 9$$

Question: $y = \tan^{-1} \left(\frac{3 \cos x - 4 \sin x}{4 \cos x + 3 \sin x} \right) + \tan^{-1} \left(\frac{x}{1 + \sqrt{1+x^2}} \right)$, then find $\frac{dy}{dx}$ at $x = \frac{\sqrt{3}}{2}$

Options:

- (a) $\frac{5}{7}$
- (b) $\frac{-5}{7}$
- (c) $\frac{3}{7}$
- (d) $\frac{-3}{7}$

Answer: (b)

Solution:

$$y_1 = \tan^{-1} \left(\frac{\frac{3}{4} - \tan x}{1 + \frac{3}{4} \tan x} \right)$$

$$\tan^{-1} \left(\frac{A-B}{1+AB} \right) = \tan^{-1} A - \tan^{-1} B, \text{ let } A = \frac{3}{4} \text{ and } B = \tan x:$$

$$y_1 = \tan^{-1} \left(\frac{3}{4} \right) - \tan^{-1}(\tan x) = \tan^{-1} \left(\frac{3}{4} \right) - x$$

$$y_2 = \tan^{-1} \left(\frac{x}{1 + \sqrt{1+x^2}} \right)$$

$$\text{Let } x = \tan \theta, \text{ so } \sqrt{1+x^2} = \sec \theta:$$

$$y_2 = \tan^{-1} \left(\frac{\tan \theta}{1 + \sec \theta} \right) = \tan^{-1} \left(\frac{\frac{\sin \theta}{\cos \theta}}{1 + \frac{1}{\cos \theta}} \right) = \tan^{-1} \left(\frac{\sin \theta}{1 + \cos \theta} \right)$$

$$y_2 = \tan^{-1} \left(\frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{\theta}{2}$$

$$y_2 = \frac{1}{2} \tan^{-1} x$$

$$y = \left[\tan^{-1} \left(\frac{3}{4} \right) - x \right] + \left[\frac{1}{2} \tan^{-1} x \right]$$

$$\frac{dy}{dx} = 0 - 1 + \frac{1}{2} \cdot \frac{1}{1+x^2} = \frac{1}{2(1+x^2)} - 1$$

$$\frac{dy}{dx} = \frac{1}{2(1+\frac{3}{4})} - 1 = \frac{1}{2(\frac{7}{4})} - 1 = \frac{1}{\frac{7}{2}} - 1 = \frac{2}{7} - 1$$

$$\frac{dy}{dx} = -\frac{5}{7}$$

Question: The domain of the function $\cos^{-1} \left\{ \frac{4x + 2[x]}{3} \right\}$, where $[.]$ is greatest integer function, is

Options:

- (a) $\left[0, \frac{3}{4} \right]$
- (b) $\left[\frac{-1}{4}, \frac{1}{4} \right]$
- (c) $\left[\frac{-1}{4}, \frac{3}{4} \right]$
- (d) $\left[\frac{-3}{4}, \frac{1}{4} \right]$

Answer: (c)

Solution:

$$-1 \leq \frac{4x + 2[x]}{3} \leq 1$$

$$-3 \leq 4x + 2[x] \leq 3$$

$$\text{Let } [x] = n, \quad x \in [n, n+1)$$

$$\frac{-2n - 3}{4} \leq x \leq \frac{3 - 2n}{4}$$

$$n = -1, 0$$

$$\text{For } n = -1: x \in \left[-\frac{1}{4}, 0\right)$$

$$\text{For } n = 0: x \in \left[0, \frac{3}{4}\right]$$

$$\boxed{\left[-\frac{1}{4}, \frac{3}{4}\right]}$$

Question: Let S_n is the sum of first n terms of an A.P. and a_n is the n^{th} term of A.P. If first term of the given A.P is $\frac{10}{3}$ and $S_{30} = a_{30}^3$. Then, the common difference of the given A.P is

Solution:

$$S_{30} = 15 \left[2 \cdot \frac{10}{3} + 29d \right] = 15 \left[\frac{20}{3} + 29d \right]$$

$$a_{30} = \frac{10}{3} + 29d$$

$$15 \left(\frac{20}{3} + 29d \right) = \left(\frac{10}{3} + 29d \right)^3$$

$$x = \frac{10}{3} + 29d$$

$$15 \left(x - \frac{10}{3} + \frac{20}{3} \right) = x^3$$

$$15 \left(x + \frac{10}{3} \right) = x^3$$

$$15x + 50 = x^3$$

$$x^3 - 15x - 50 = 0$$

$$x = 5 \Rightarrow 125 - 75 - 50 = 0$$

$$\frac{10}{3} + 29d = 5$$

$$29d = 5 - \frac{10}{3} = \frac{5}{3}$$

$$d = \frac{5}{87}$$

Question: The value of $A = \frac{\sin 3^\circ}{\cos 9^\circ} + \frac{\sin 9^\circ}{\cos 27^\circ} + \frac{\sin 27^\circ}{\cos 81^\circ}$ and $B = \tan 81^\circ - \tan 3^\circ$. The value of $\frac{A}{B}$ is equal to

Solution:

$$\text{Term 1: } \frac{\sin 3^\circ}{\cos 9^\circ} = \frac{1}{2}(\tan 9^\circ - \tan 3^\circ)$$

$$\text{Term 2: } \frac{\sin 9^\circ}{\cos 27^\circ} = \frac{1}{2}(\tan 27^\circ - \tan 9^\circ)$$

$$\text{Term 3: } \frac{\sin 27^\circ}{\cos 81^\circ} = \frac{1}{2}(\tan 81^\circ - \tan 27^\circ)$$

$$A = \frac{1}{2} [(\tan 9^\circ - \tan 3^\circ) + (\tan 27^\circ - \tan 9^\circ) + (\tan 81^\circ - \tan 27^\circ)]$$

$$A = \frac{1}{2} (\tan 81^\circ - \tan 3^\circ)$$

$$\frac{A}{B} = \frac{1}{2}$$

Question: Let $P(1, 2)$ be any point from which two chords bisected at y -axis are drawn to the circle whose equation is $x^2 + y^2 + x - 6y = 0$. Then, a equation of the chord is

Options:

- (a) $(1 + \sqrt{3})x + 2y - (5 + \sqrt{3}) = 0$
- (b) $(1 + \sqrt{3})x - 2y + (5 + \sqrt{3}) = 0$
- (c) $(1 - \sqrt{3})x + 2y + (5 + \sqrt{3}) = 0$
- (d) $(1 - \sqrt{3})x - 2y + (5 + \sqrt{3}) = 0$

Answer: (a)

Solution:

1. Identify the Circle Properties

The given circle equation is:

$$x^2 + y^2 + x - 6y = 0$$

2. Define the Midpoint

The problem states that the chord is **bisected at the y -axis**. This means its midpoint, let's call it M , must lie on the y -axis. Therefore, the coordinates of the midpoint are $M(0, y_1)$.

3. Use the Chord Equation Formula ($T = S_1$)

The equation of a chord of a circle with a known midpoint (x_1, y_1) is given by $T = S_1$.
For our circle and midpoint $M(0, y_1)$:

- **T (the tangent expression):**

$$x(0) + y(y_1) + \frac{1}{2}(x+0) - 3(y+y_1) = \frac{x}{2} + (y_1-3)y - 3y_1$$

- **S_1 (the power of the point):** $0^2 + y_1^2 + 0 - 6y_1 = y_1^2 - 6y_1$

Setting $T = S_1$:

$$\frac{x}{2} + (y_1-3)y - 3y_1 = y_1^2 - 6y_1$$

$$\frac{x}{2} + (y_1-3)y = y_1^2 - 3y_1 \quad \text{--- (Equation 1)}$$

4. Solve for the Midpoint Coordinate (y_1)

The chord passes through the point **P(1,2)**. Substituting $x = 1$ and $y = 2$ into Equation 1:

$$\frac{1}{2} + 2(y_1-3) = y_1^2 - 3y_1$$

$$0.5 + 2y_1 - 6 = y_1^2 - 3y_1$$

$$y_1^2 - 5y_1 + 5.5 = 0$$

Multiply by 2 to remove decimals:

$$2y_1^2 - 10y_1 + 11 = 0$$

Using the quadratic formula $y_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$y_1 = \frac{10 \pm \sqrt{100 - 88}}{4} = \frac{10 \pm \sqrt{12}}{4} = \frac{5 \pm \sqrt{3}}{2}$$

5. Final Chord Equations

By substituting these two values of y_1 back into Equation 1, we get the two possible chords:

- **Case 1:** $y_1 = \frac{5 + \sqrt{3}}{2}$

$$x + (\sqrt{3} - 1)y + (1 - 2\sqrt{3}) = 0$$

- **Case 2:** $y_1 = \frac{5 - \sqrt{3}}{2}$

$$x - (1 + \sqrt{3})y + (1 + 2\sqrt{3}) = 0$$

Question: Let $f(x) = f'(x)$, $f'(X)$ and $f(0) = 0$, where $f(x)$ is a polynomial function of n^{th} degree. The value of $\left[f'(2) + f''(2) + \int_0^1 f(x) dx \right]$ is equal to

Options:

- (a) 36
- (b) 40
- (c) 48.5
- (d) 38.5

Answer: (d)

Solution:

1. Find the degree of the polynomial

Let $f(x)$ be a polynomial of degree n .

- The degree of its first derivative $f'(x)$ is $n - 1$.
- The degree of its second derivative $f''(x)$ is $n - 2$.

The given equation is $f(x) = f'(x) \cdot f''(x)$. Comparing the degrees on both sides:

$$n = (n - 1) + (n - 2)$$

$$n = 2n - 3 \implies n = 3$$

So, $f(x)$ is a **cubic polynomial**.

2. Determine the polynomial $f(x)$

Let $f(x) = ax^3 + bx^2 + cx + d$.

Since $f(0) = 0$, we have $d = 0$. Thus, $f(x) = ax^3 + bx^2 + cx$.

Now, find the derivatives:

- $f'(x) = 3ax^2 + 2bx + c$
- $f''(x) = 6ax + 2b$

Substitute these into $f(x) = f'(x) \cdot f''(x)$:

$$ax^3 + bx^2 + cx = (3ax^2 + 2bx + c)(6ax + 2b)$$

$$ax^3 + bx^2 + cx = 18a^2x^3 + (6ab + 12ab)x^2 + (4b^2 + 6ac)x + 2bc$$

Comparing the coefficients of like powers of x :

- **x^3 term:** $a = 18a^2 \implies a = \frac{1}{18}$ (since $a \neq 0$ for a cubic).
- **Constant term:** $0 = 2bc$. This implies $b = 0$ or $c = 0$.
- **x^2 term:** $b = 18ab \implies b = 18(\frac{1}{18})b$, which is always true.

- **x term:**

$$c = 4b^2 + 6ac \implies c = 4b^2 + 6\left(\frac{1}{18}\right)c \implies c = 4b^2 + \frac{1}{3}c \implies \frac{2}{3}c = 4b^2 \implies c = 6b^2$$

From $2bc = 0$ and $c = 6b^2$, we get $2b(6b^2) = 12b^3 = 0 \implies \mathbf{b = 0}$.

Then $c = 6(0)^2 = 0$.

So, the polynomial is:

$$f(x) = \frac{1}{18}x^3$$

3. Evaluate the components of the expression

- **$f'(2)$:** $f'(x) = \frac{3}{18}x^2 = \frac{1}{6}x^2 \implies f'(2) = \frac{1}{6}(2)^2 = \frac{2}{3}$

- **Integral:** $\int_0^1 f(x)dx = \int_0^1 \frac{1}{18}x^3dx = \left[\frac{x^4}{72}\right]_0^1 = \frac{1}{72}$

- **$f''(x)$:** $f''(x) = \frac{6}{18}x = \frac{1}{3}x$.

4. Final Calculation

Assuming the expression is evaluated at $x = 2$:

$$\text{Value} = 36 \left[f'(2) + f''(2) + \int_0^1 f(x)dx \right]$$

$$\text{Value} = 36 \left[\frac{2}{3} + \frac{2}{3} + \frac{1}{72} \right]$$

$$\text{Value} = 36 \left[\frac{4}{3} + \frac{1}{72} \right] = 36 \left(\frac{4}{3} \right) + 36 \left(\frac{1}{72} \right)$$

$$\text{Value} = 48 + 0.5 = \mathbf{48.5}$$

Question: If the value of $\int_{-2}^2 (|\sin x| + |x \sin x|)dx = \beta(1 - 3 \cos \alpha) + \beta \sin \beta$ then β is

equal to

Options:

- (a) 3
- (b) 2
- (c) 5
- (d) -2

Answer: (b)

Solution:

1. Analyze the Integrand

The integrand is $f(x) = |\sin x| + |x \sin x|$.

We can factor out $|\sin x|$:

$$f(x) = |\sin x|(1 + |x|)$$

Notice that $f(x)$ is an **even function** because:

- $|\sin(-x)| = |-\sin x| = |\sin x|$
- $|(-x)| = |x|$
- Therefore, $f(-x) = f(x)$.

Using the property of even functions for symmetric limits $\int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx$, the integral becomes:

$$I = 2 \int_0^2 |\sin x|(1 + x)dx$$

2. Simplify for the Interval $[0, 2]$

In the interval $x \in [0, 2]$, $\sin x$ is always positive (since $2 < \pi \approx 3.14$). Thus, $|\sin x| = \sin x$.

$$I = 2 \int_0^2 (\sin x + x \sin x)dx$$

$$I = 2 \left[\int_0^2 \sin x dx + \int_0^2 x \sin x dx \right]$$

3. Evaluate the Integrals

- **First Part:**

$$\int_0^2 \sin x dx = [-\cos x]_0^2 = -\cos 2 - (-\cos 0) = 1 - \cos 2$$

- **Second Part (using Integration by Parts):**

Let $u = x$ and $dv = \sin x dx$. Then $du = dx$ and $v = -\cos x$.

$$\int x \sin x dx = -x \cos x - \int (-\cos x) dx = -x \cos x + \sin x$$

Evaluating from 0 to 2:

$$[-2 \cos 2 + \sin 2] - [-0 \cos 0 + \sin 0] = \sin 2 - 2 \cos 2$$

4. Combine and Compare

Now, sum the parts and multiply by 2:

$$I = 2[(1 - \cos 2) + (\sin 2 - 2 \cos 2)]$$

$$I = 2[1 - 3 \cos 2 + \sin 2]$$

$$I = 2(1 - 3 \cos 2) + 2 \sin 2$$

Comparing this result with the given form:

$$\beta(1 - 3 \cos \alpha) + \beta \sin \beta$$

We can clearly see that $\beta = 2$ (and $\alpha = 2$).

Question: If Latus rectum of parabola $y^2 = 4kx$ and ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ coincide then the value of $e^2 + 2\sqrt{2}$ is, where e is eccentricity of ellipse

Options:

- (a) 2
- (b) 3
- (c) 4
- (d) 3/2

Answer: (b)

Solution:

1. Properties of the Latus Rectum (LR)

• For the Parabola $y^2 = 4kx$:

- The focus is at $(k, 0)$.
- The equation of the LR is $x = k$.
- The length of the LR is $4k$. Therefore, the endpoints are $(k, 2k)$ and $(k, -2k)$.

• For the Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

- The focus is at $(ae, 0)$.
- The equation of the LR is $x = ae$.
- The length of the LR is $\frac{2b^2}{a}$. Therefore, the endpoints are $(ae, \frac{b^2}{a})$ and $(ae, -\frac{b^2}{a})$

2. The "Coincide" Condition

For the two line segments to coincide, their equations and their endpoints must be identical:

1. From the equations: $k = ae$

2. From the endpoints: $2k = \frac{b^2}{a}$

3. Solving for Eccentricity (e)

Substitute the value of k from the first equation into the second:

$$2(ae) = \frac{b^2}{a} \implies 2a^2e = b^2$$

We know the standard relation for the eccentricity of an ellipse is $b^2 = a^2(1 - e^2)$.

Substituting this for b^2 :

$$2a^2e = a^2(1 - e^2)$$

Dividing both sides by a^2 (since $a \neq 0$):

$$2e = 1 - e^2$$

$$e^2 + 2e - 1 = 0$$

Solving this quadratic equation using the formula $e = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$e = \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2} = \frac{-2 \pm \sqrt{8}}{2} = \frac{-2 \pm 2\sqrt{2}}{2} = -1 \pm \sqrt{2}$$

Since the eccentricity e of an ellipse must be a positive value between 0 and 1, we choose:

$$e = \sqrt{2} - 1$$

4. Final Calculation

First, let's find e^2 :

$$e^2 = (\sqrt{2} - 1)^2 = 2 + 1 - 2\sqrt{2} = 3 - 2\sqrt{2}$$

Now, substitute this into the required expression:

$$\text{Value} = e^2 + 2\sqrt{2}$$

$$\text{Value} = (3 - 2\sqrt{2}) + 2\sqrt{2}$$

$$\text{Value} = 3$$

Question: If $\frac{dy}{dx} = (1 + x + x^2)(1 - y + y^2)$, $y(0) = \frac{1}{2}$ then $2y(1) - 1$ is

Solution:

1. Separate the Variables

We group all terms involving y on the left and all terms involving x on the right:

$$\frac{dy}{1-y+y^2} = (1+x+x^2)dx$$

2. Integrate Both Sides

Now, we integrate both sides of the equation:

$$\int \frac{1}{y^2 - y + 1} dy = \int (1 + x + x^2) dx$$

- **Left Side (LHS):** We complete the square for the denominator.

$$y^2 - y + 1 = (y - \frac{1}{2})^2 + \frac{3}{4} = (y - \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2$$

The integral follows the form $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1}(\frac{u}{a})$:

$$\text{LHS} = \frac{1}{\sqrt{3}/2} \tan^{-1}\left(\frac{y-1/2}{\sqrt{3}/2}\right) = \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)$$

- **Right Side (RHS):**

$$\text{RHS} = x + \frac{x^2}{2} + \frac{x^3}{3} + C$$

Combining them:

$$\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right) = x + \frac{x^2}{2} + \frac{x^3}{3} + C$$

3. Use the Initial Condition

We are given that $y(0) = \frac{1}{2}$. Substitute $x = 0$ and $y = \frac{1}{2}$ to find C :

$$\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2(1/2)-1}{\sqrt{3}}\right) = 0 + 0 + 0 + C$$

$$\frac{2}{\sqrt{3}} \tan^{-1}(0) = C \implies C = 0$$

So, the general solution is:

$$\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right) = x + \frac{x^2}{2} + \frac{x^3}{3}$$

4. Find the Value of $2y(1) - 1$

Substitute $x = 1$ into the equation:

$$\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2y(1)-1}{\sqrt{3}}\right) = 1 + \frac{1}{2} + \frac{1}{3}$$

$$\frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2y(1)-1}{\sqrt{3}}\right) = \frac{11}{6}$$

Isolate the $\tan^{-1}$ term:

$$\tan^{-1}\left(\frac{2y(1)-1}{\sqrt{3}}\right) = \frac{11}{6} \cdot \frac{\sqrt{3}}{2} = \frac{11\sqrt{3}}{12}$$

Taking the tangent of both sides:

$$\frac{2y(1)-1}{\sqrt{3}} = \tan\left(\frac{11\sqrt{3}}{12}\right)$$

$$2y(1)-1 = \sqrt{3} \tan\left(\frac{11\sqrt{3}}{12}\right)$$

Question: A coin is tossed 8 times. Find probability that there are 4 heads in first 6 and 3 heads in last 5 tosses.

Solution:

1. Total Possible Outcomes

When a coin is tossed 8 times, each toss has 2 possibilities (Heads or Tails).

Total outcomes = $2^8 = 256$

2. Analyzing the Constraints

Let the 8 tosses be represented as $T_1, T_2, T_3, T_4, T_5, T_6, T_7, T_8$.

- **Condition 1:** 4 heads in the first 6 tosses (T_1 to T_6).
- **Condition 2:** 3 heads in the last 5 tosses (T_4 to T_8).

Notice that the tosses T_4, T_5, T_6 are common to both conditions. Let's group the tosses to handle the overlap:

- **Group A:** $\{T_1, T_2, T_3\}$ (3 tosses)
- **Group B (Overlap):** $\{T_4, T_5, T_6\}$ (3 tosses)

- **Group C:** $\{T_7, T_8\}$ (2 tosses)

Let h_A, h_B , and h_C be the number of heads in each group. Our conditions translate to:

1. $h_A + h_B = 4$
2. $h_B + h_C = 3$

3. Calculating Successful Cases

Since h_B can only be 1, 2, or 3 (it must be at least 1 for h_A to be at most 3, and at least 1 for h_C to be at most 2), we check these cases:

Case	h_B (Heads in B)	h_A (Must be $4 - h_B$)	h_C (Must be $3 - h_B$)	Ways to arrange (nC_r)	Sub-total
1	1 head	3 heads	2 heads	${}^3C_3 \times {}^3C_1 \times {}^2C_2 = 1 \times 3 \times 1$	3
2	2 heads	2 heads	1 head	${}^3C_2 \times {}^3C_2 \times {}^2C_1 = 3 \times 3 \times 2$	18
3	3 heads	1 head	0 heads	${}^3C_1 \times {}^3C_3 \times {}^2C_0 = 3 \times 1 \times 1$	3

Total successful outcomes = $3 + 18 + 3 = 24$

4. Final Probability

The probability (P) is the ratio of successful outcomes to total outcomes:

$$P = \frac{24}{256}$$

By dividing both numbers by 8, we simplify the fraction:

$$P = \frac{3}{32}$$

In decimal form, this is **0.09375**.